CS683 PA2: Dynamic I-RRIP

Trace1.gz; single-core ChampSim; 25 million warmup and 25 million simulation instructions.

Methodology

Policy design: Dynamic I-RRIP uses set dueling at L2. Sets 0–15 are fixed SRRIP leader sets, sets 16–31 are fixed I-RRIP leader sets, and all remaining sets are follower sets controlled by the selector.

Selector statistic: Counters are updated for accesses to either group of leader sets. Let I_A and I_M be instruction accesses and misses, and A and M be total accesses and misses in those leader sets. The score is

$$\text{score} = (I_A + I_M) / (A + M)$$

The selector chooses I-RRIP for follower sets when score exceeds the configured threshold; otherwise it chooses SRRIP. The default threshold is 0.25, and the experiment varies it across five values.

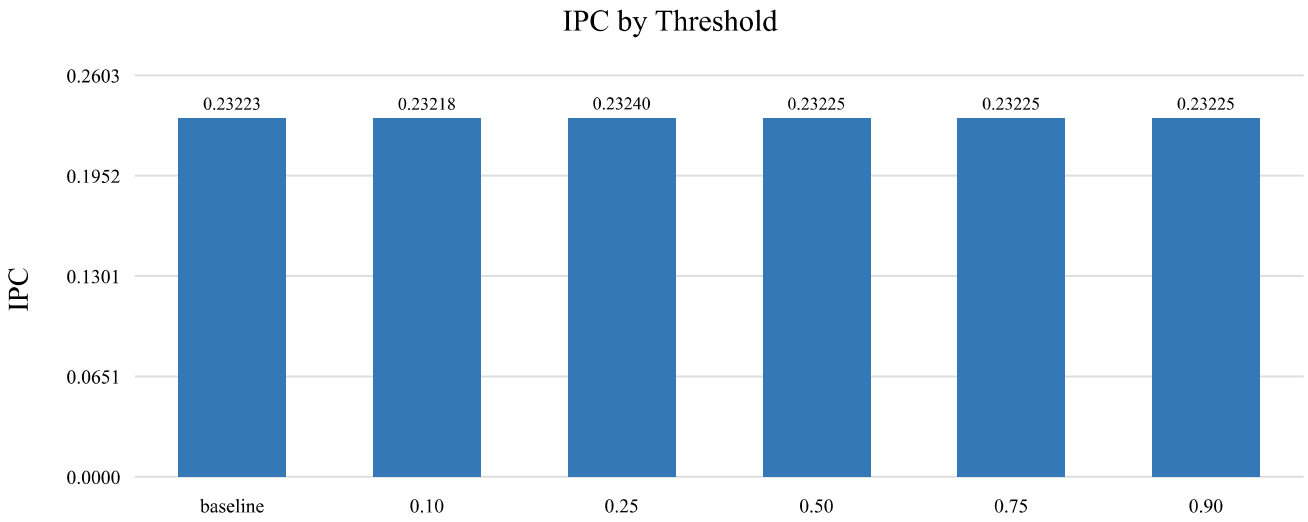
Insertion and victim selection: Hits reset the accessed line's RRPV to 0. On a miss, the implementation inserts instruction lines at RRPV 2. Under I-RRIP, data lines are inserted at RRPV 3; otherwise, lines are inserted at RRPV 2. Victim selection chooses a line at RRPV 3, increasing RRPVs when necessary until a victim is available.

Experimental setup: Single-core ChampSim, Trace1.gz, 25 million warmup instructions followed by 25 million simulation instructions. The results represent this trace and configuration only.

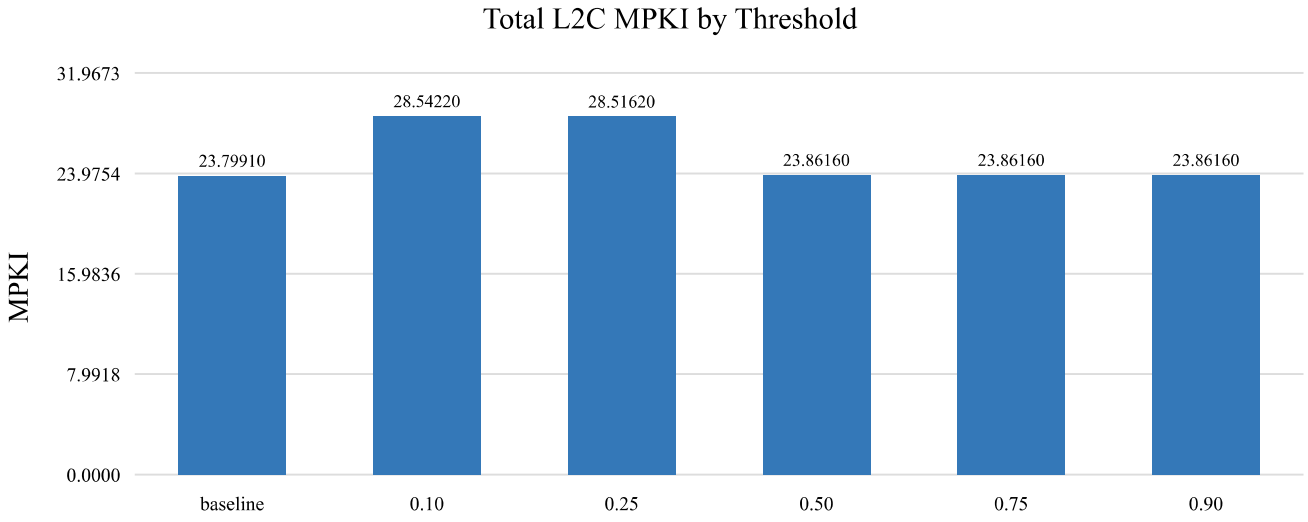
Experimental results

Threshold	IPC	Speedup	Total MPKI	Data-load MPKI	Instruction-load MPKI	Follower policy
baseline	0.232232	1.000000x	23.7991	7.06972	10.5255	SRRIP
0.10	0.232178	0.999767x	28.5422	8.67768	9.66028	I-RRIP
0.25	0.232396	1.000706x	28.5162	8.65824	9.6566	I-RRIP
0.50	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP
0.75	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP
0.90	0.232250	1.000078x	23.8616	7.09548	10.5098	RRIP

IPC by threshold



Total L2C MPKI by threshold



Discussion

Baseline SRRIP IPC is 0.232232. Threshold 0.25 has IPC 0.232396, but total L2C MPKI rises from 23.7991 to 28.5162. Instruction-load MPKI falls while data-load MPKI rises, indicating a trade-off rather than a substantial overall performance improvement.

Thresholds 0.50, 0.75 and 0.90 have identical measured results and finish with RRIP as the follower policy. Thresholds 0.10 and 0.25 finish with I-RRIP. Only Trace1 was available for this sweep, so results should not be generalized to other workloads.